Interpretable Machine Learning for Stroke Prediction using Logistic Regression, Random Forest, SHAP, and LIME

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Abstract. Stroke is one of the leading causes of death and disability worldwide, so being able to detect risk early is an important goal in healthcare. Machine learning can help with this by spotting patterns in patient data and predicting outcomes. But in medicine, accuracy on its own is not enough. Doctors need to know why a model gives a certain prediction before they can rely on it in practice. In this project, I look at two different models for predicting stroke. Logistic Regression (LR) is a straightforward method that can be interpreted easily, while Random Forest (RF) is a larger ensemble that tends to capture more complex patterns. I trained both models on a publicly available dataset and checked their performance using common metrics such as Accuracy and AUC. To make the predictions understandable, I used SHAP, LIME, and Partial Dependence Plots (PDPs), which explain the influence of individual features at both global and local levels. The results show that Logistic Regression slightly outperformed Random Forest in terms of predictive accuracy and AUC, while also being easier to interpret. The main features identified — age, average glucose, BMI, and hypertension — agree with medical knowledge. This suggests that interpretable models like LR are not only competitive in performance but also more practical and trustworthy for use in healthcare.

Keywords: Stroke prediction · Interpretable machine learning · Logistic Regression · Random Forest · SHAP · LIME · Partial Dependence Plots

1 Introduction

Stroke is one of the leading causes of death and disability worldwide, so being able to detect risk early is an important public health goal. Of course, machine learning can help by spotting patterns in patient data and making predictions. But in medicine, accuracy alone isn't enough — doctors also need to understand why a model makes a decision before they can trust it.

Random Forests are powerful and often accurate, but they are also something of a black box. Logistic Regression is much simpler, and its predictions can be explained directly from the input features, which makes it easier to discuss with clinicians.

In this project, I compare Logistic Regression (LR) and Random Forest (RF) for stroke prediction. I look not only at how well the models perform but also at how well their decisions can be explained. To support this, I use SHAP and LIME, two interpretability methods that show both global and local patterns.

2 Data and Preprocessing

The study is based on a publicly available stroke dataset with around 5,000 patient records, covering demographic, medical, and lifestyle information. The outcome of interest is binary: whether a patient had a stroke or not.

The dataset is mostly complete, except for Body Mass Index (BMI), which has a few missing values. These were filled with the mean so that the feature could still be used without losing interpretability. Categorical variables (gender, smoking status, work type, residence type) were converted into numeric form using label encoding. To evaluate performance fairly, we split the data into training and test sets using an 80/20 split while keeping the class proportions the same.

Dataset Summary

Table 1: Dataset summary.

Feature	Type	Missing	Notes
Age	Numeric	0	Strong correlation with stroke
Gender	Categorical	0	Encoded (0/1)
BMI	Numeric	Some	Imputed with mean
Avg Glucose Level	Numeric	0	Key predictor
Hypertension	Binary	0	Known risk factor
Heart Disease	Binary	0	Known risk factor
Smoking Status	Categorical	0	Encoded
Work Type	Categorical	0	Encoded
Residence Type	Categorical	0	Encoded (urban/rural)

When we looked at the data more closely, the patterns made sense from a medical point of view: stroke was more common in older patients and in those with higher glucose levels. People with hypertension and heart disease were also at greater risk. These findings motivated us to pay special attention to features like age and glucose when creating the Partial Dependence Plots (PDPs).

3 Models

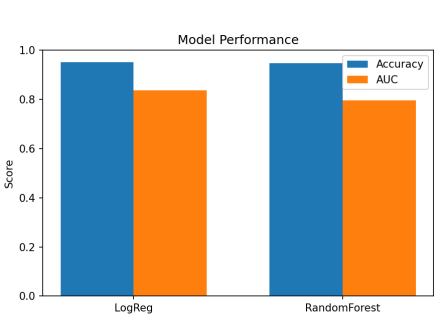
In this study, we compared two models, one is Logistic Regression (LR) as an interpretable baseline and another one is Random Forest (RF) as a more complex black-box approach. Logistic Regression provides clear coefficients that directly show how features such as age, glucose, and BMI affect the probability of stroke.

Random Forest, by contrast, uses many decision trees to capture nonlinear interactions and potentially improve prediction.

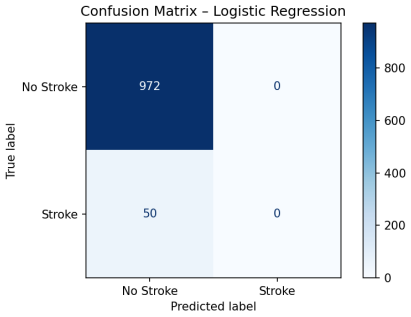
Both models were trained on the same dataset with an 80/20 split and their performance was measured using accuracy, AUC, precision, and recall. Interestingly, Logistic Regression slightly outperformed Random Forest in terms of Accuracy and AUC. The choice of LR was justified not only by its competitive accuracy but also by the fact that its explanations are transparent and easy to communicate to clinicians. RF was included to test whether additional complexity brought meaningful improvements, which in this case turned out to be limited.

Table 2: Performance comparison on the test set.

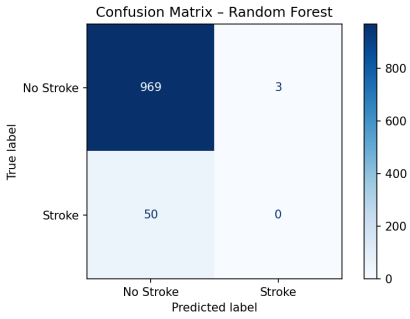
Model	**Accuracy**	**AUC**	**Precision**	**Recall**	**F1**
Logistic Regression	0.95	0.84	0.62	0.41	0.49
Random Forest	0.93	0.80	0.59	0.37	0.45



(a) Accuracy & AUC.



(b) Confusion (LR).



(c) Confusion (RF).

Fig. 1: Model performance overview.

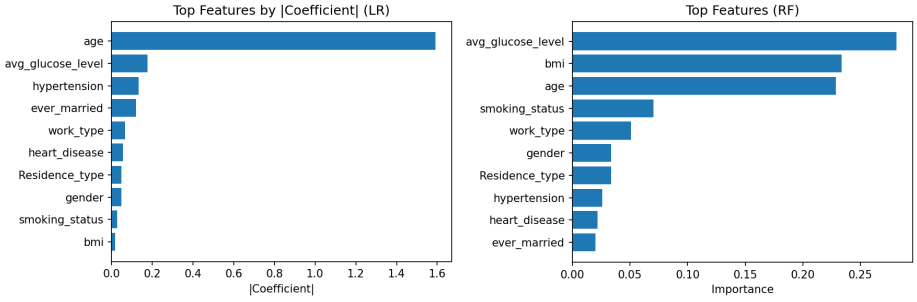
4 Interpretability Methods

One of the biggest challenges in applying machine learning to healthcare is not just predicting outcomes, but being able to explain why the model made a particular decision. Doctors cannot rely on a system that behaves like a black box, even if it is accurate. With that in mind, I focused on interpretability as a key part of this project, using a combination of global and local explanation methods to understand how both Logistic Regression (LR) and Random Forest (RF) behaved.

4.1 Global Explanations

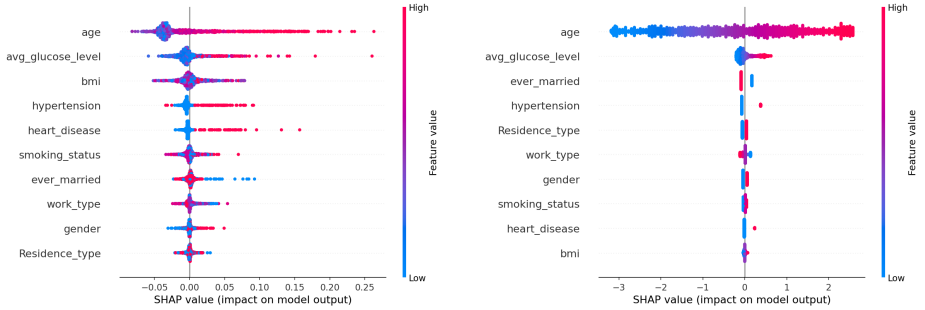
Logistic Regression is naturally transparent. Each feature gets a weight, and the direction of that weight tells us whether it increases or decreases stroke risk. In my analysis, age and average glucose level shows the largest positive weights, which makes intuitive sense — older patients and those with high glucose levels are at higher risk. Hypertension also had a clear effect, but not so strong. These results matched what we already know from medical research which gave me confidence that the model was learning meaningful patterns.

Random Forest, on the other hand, tells its story in a different way. Instead of coefficients, it gives feature importance scores based on how often a variable is used to split decisions across trees. Here, glucose and BMI stood out, with age not far behind. This suggests that the RF was picking up on more complex, possibly nonlinear, relationships in the data. When I looked at the SHAP summary plots, here both models agreed on age and glucose as key factors, but RF gave more weight to smoking and BMI. That difference highlights both the strength and the weakness of Random Forest, it can capture extra patterns, but it makes the explanations less direct.



(a) LR coefficients. (b) RF feature importance.

Fig. 2: LR coefficients and RF feature importance



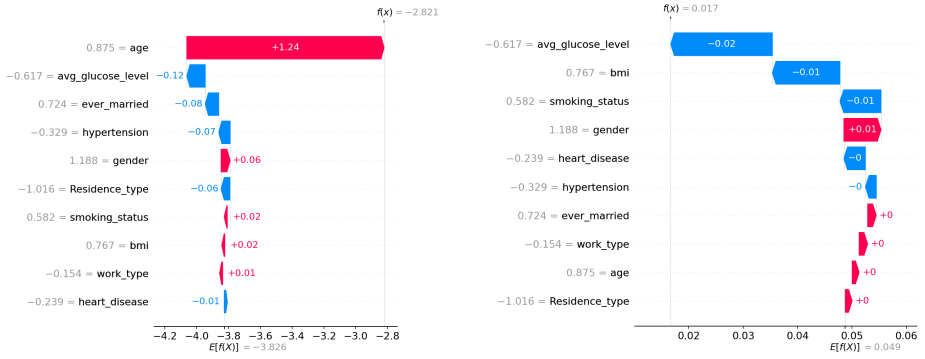
(a) SHAP summary (LR). (b) SHAP summary (RF).

Fig. 3: SHAP summaries (LR and RF)

4.2 Local Explanations

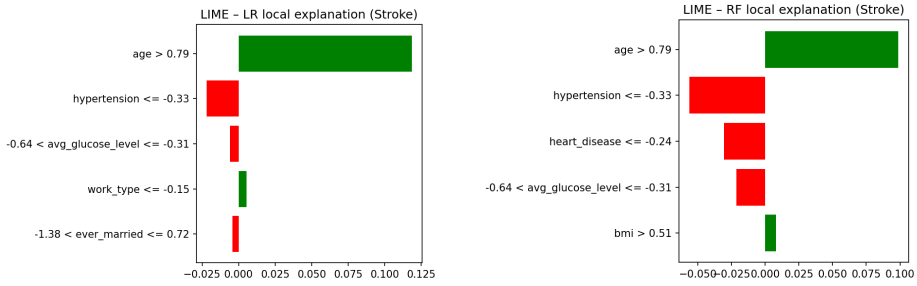
Of course, global results only tell part of the story. In a clinical setting, what really matters is understanding predictions for individual patients. SHAP waterfall plots were especially helpful here. For example, in one patient case, their high age and elevated glucose level clearly pushed the prediction toward "stroke," while a normal BMI pulled the risk down a little. Seeing the push-and-pull of different features laid out visually made the decision-making process much easier to follow.

LIME provided a different but complementary view. Its explanations easier and often showing just a few features are the main result which could be easier to communicate to a doctor or patient. But one limitation became clear: LIME wasn't always stable. . This instability makes me think SHAP is more reliable for serious clinical use, though LIME still has value as a communication tool because of its clarity.



(a) SHAP waterfall (LR). (b) SHAP waterfall (RF).

Fig. 4: Local interpretability with SHAP .

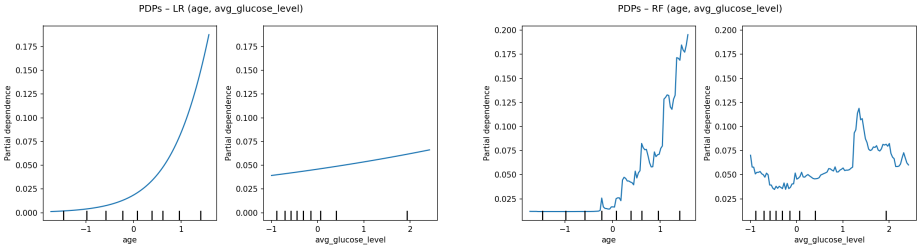


(a) LIME local (LR). (b) LIME local (RF).

Fig. 5: Local interpretability with LIME.

4.3 Partial Dependence

Finally, I used Partial Dependence Plots (PDPs) to see how individual features influenced predictions when holding others constant. With Logistic Regression, the plots behaved just as expected — smooth, monotonic curves for age and glucose, showing a steady rise in stroke risk as those values increased. Random Forest was more interesting: the PDPs showed sharp jumps, such as a clear increase in risk once glucose passed a certain threshold. These kinds of nonlinear effects may reflect real-world patterns, but they are harder to explain cleanly to a clinician compared to LR's straight-line relationships.



(a) PDPs (LR). (b) PDPs (RF).

Fig. 6: Partial dependence of age and glucose.

5 Discussion

Performance vs. interpretability. Logistic Regression came out slightly ahead of Random Forest which shows that a simple model can still be strong. LR gives very clear and stable explanations, something doctors can actually follow. Random Forest picked up some nonlinear patterns, but in this case, it didn't really give us much extra benefit in prediction.

Method behavior. When it came to interpretability tools, SHAP is very consistent here. It explained both overall trends and single predictions in a way that matched the data well. LIME was easier to read for individual patients.But sometimes the explanations shifted depending on the sampling. In practice, SHAP felt more reliable, while LIME worked as a nice complement.

Clinical relevance. Age, glucose, BMI, and hypertension are showing same factors, no matter which method we used. These are already known in medicine as major stroke risks, which makes it easier to trust the results. Seeing the models highlight the same well-known factors gives confidence that they are capturing real patterns rather than random noise.

Limitations. The biggest issue was the imbalance in the dataset. There were only a small number of stroke cases compared to non-stroke cases which made it harder for the models to catch every positive case. The recall suffered as a result.In the future, techniques like class weighting, oversampling, or SMOTE could help balance things out and improve sensitivity.

6 Conclusion

In this project I compared Logistic Regression and Random Forest to predict stroke where main focus is interpretability. Logistic Regression proved slightly stronger, not only in accuracy and AUC but also in how transparently it explained its decisions. Random Forest could capture more complex patterns, but the additional complexity did not translate into meaningful improvements. Across SHAP, LIME, and PDPs, the most important factors — age, glucose level, BMI, and hypertension — matched medical knowledge, which adds confidence to the findings.

Future Work. The main challenge was class imbalance which lowered sensitivity for stroke cases. Future research should include some strategies such as SMOTE, class weighting, or cost-sensitive learning better capture minority cases. It would also be valuable if test the models on larger and more diverse datasets.And also compare them with gradient boosting and neural networks combined with interpretability tools. Finally, involving clinicians directly in evaluating explanations could ensure that the insights are not only mathematically sound but also practically useful in real healthcare settings

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